CMSC3621 Data Structures and Algorithms Lab

Lab Assignment 1

Overview:

This assignment is for students to review the usage of struct, recursion, and STL (i.e., queue, and stack) in C++. The students simply need to practice four C++ programs.

Requirements:

1. Complete programs in C++.

2. Submit the source code and a document including screenshots and explanations to D2L. (Please see the instructions and example in the end of this document)

Instructions:

Q1 . Please define 3 structure type variables `struct1`, `struct2`, and `struct3`.

The type of `struct3` is **S3**. There are two member variables in the **S3** structure:

- `int mv5`
- `string mv6`

The type of `struct2` is **S2**. There are two member variables in the **S2** structure:

- `struct3 * mv3 // A pointer that points to a variable whose type is struct3`
- `int mv4`

The type of `struct1` is **S1**. There are two member variables in the **S1** structure:

- `string mv1`
- `struct2 mv2  // The data type of mv2 is struct2`

2) Initialize the three variables `struct1`, `struct2`, and `struct3`.

3) Access the value of `mv6` through `struct1`, and output the value.

Q2

Implement a C++ program which can determine whether the parentheses in any given string `s` are used correctly or not. You must use STL stack adapter.

Some examples:

- `)(`
- `((())`
- `()`
- `(a)( (b)((c)(d )))`

The last two are valid, but the first two are not.

Q3

Complete the following C++ program which is used to find the smallest value from an array. Please note you are NOT allowed to use any ‘loop’. Instead, you must use recursion.

```
#include <iostream>
#include <cstdlib>
using namespace std;

int find_smallest(int a[], int size) {
// To do
}

int main(int argc, char* argv[]) {
     int arr[] = {13, 19, 12, 11, 15, 19, 23, 12,
                  13, 22, 18, 19, 14, 17, 23, 21};
     cout<< "smallest: " << find_smallest(arr, 16) << endl;
}
```

Q4

Implement a C++ program which can print the Nth row in a Pascal's Triangle (https://en.wikipedia.org/wiki/Pascal%27s_triangle).

You must use the command-line parameter to read an input.

For example,

if the input number is 3, the output should be 1, 2, 1.

If the input number is 4, the output should be 1, 3, 3, 1.

You must use STL queue adapter.

Submission:

Write a report in a PDF format. Screenshots and explanations are included to show how you tested your solutions. For the second program (Q2), please make sure at least all the provided test cases are evaluated.

For the final program (Q3), please also test your programs using at least three different cases.

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1. Name the first program as Lab1_Q1.cpp;

2. Name the second program as Lab1_Q2.cpp;

3. Name the third program as Lab1_Q3.cpp;

4. Name the fourth program as Lab1_Q4.cpp;

5. Name the report as Lab1_report.pdf;

6. Submit all three files above to D2L.

Rubrics:

Q1 (20 points); Q2 (20 points); Q3 (20 points); Q4 (20 points); Report quality (20 points).

Any violation against the rules above will lead to extra penalty.

If you missed the report, your submission will not be evaluated.